1. SQL Queries for Analytical Questions

Table Names:

```
policy_sales
claims_data
```

Q1. Total Premium Collected During 2024

```
SELECT
SUM(Premium) AS Total_Premium_Collected
FROM policy_sales
WHERE YEAR(Policy_Purchase_Date) = 2024;
```

Result
₹240,022,200

Q2. Total Claim Cost per Year with Monthly Breakdown

```
SELECT
YEAR(Claim_Date) AS Claim_Year,
MONTH(Claim_Date) AS Claim_Month,
SUM(Claim_Amount) AS Total_Claim_Cost
FROM claims_data
GROUP BY YEAR(Claim_Date), MONTH(Claim_Date)
ORDER BY Claim_Year, Claim_Month;
```

This produces a **monthly claims trend for 2025 and 2026**.

Q3. Claim Cost to Premium Ratio by Policy Tenure

```
SELECT
p.Policy_Tenure,
```

```
SUM(c.Claim_Amount) / SUM(p.Premium) AS Claim_to_Premium_Ratio
FROM policy_sales p
LEFT JOIN claims_data c
ON p.Customer_ID = c.Customer_ID
AND p.Vehicle_ID = c.Vehicle_ID
GROUP BY p.Policy_Tenure
ORDER BY p.Policy_Tenure;
```

Q4. Claim Cost to Premium Ratio by Policy Sale Month

```
SELECT
MONTH(p.Policy_Purchase_Date) AS Sale_Month,
SUM(c.Claim_Amount) / SUM(p.Premium) AS Claim_to_Premium_Ratio
FROM policy_sales p
LEFT JOIN claims_data c
ON p.Customer_ID = c.Customer_ID
AND p.Vehicle_ID = c.Vehicle_ID
GROUP BY MONTH(p.Policy_Purchase_Date)
ORDER BY Sale_Month;
```

Q5. Estimate Total Potential Claim Liability

```
SELECT
COUNT(*) AS Vehicles_Without_Claim,
COUNT(*) * 10000 AS Estimated_Liability
FROM policy_sales p
LEFT JOIN claims_data c
ON p.Customer_ID = c.Customer_ID
AND p.Vehicle_ID = c.Vehicle_ID
WHERE c.Vehicle_ID IS NULL;
```

Result:

Vehicles without claims ≈ **950,984**

Estimated Liability:

₹9,509,840,000

Q6. Premium Earned Until Feb 28, 2026

Daily Premium formula:

```
Daily Premium = Premium / (Policy_Tenure * 365)
```

SQL approximation:

```
SELECT
SUM(
(Premium / (Policy_Tenure * 365)) *
DATEDIFF(day, Policy_Start_Date, '2026-02-28')
) AS Earned_Premium
FROM policy_sales
WHERE Policy_Start_Date <= '2026-02-28';
```

Earned premium:

₹65,897,302

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2. Python Queries

Tools used:

- **Python**
- **Pandas**
- **NumPy**

Steps:

Step 1: Generate Policy Dataset

```
1,000,000 policies
Vehicle value = ₹100,000
```

```
Premium = ₹100 × tenure
Purchase dates distributed across 2024
```

Step 2: Assign Tenure Distribution

```
1 year → 20%
2 years → 30%
3 years → 40%
4 years → 10%
```

Step 3: Calculate Policy Dates

```
Policy_Start_Date = Purchase_Date + 365 days
Policy_End_Date = Start_Date + Tenure
```

Step 4: Simulate Claims

Rules implemented:

2025:

- Claims only from vehicles purchased on:
  - 7th
  - 14th
  - 21st
  - 28th
- 30% of those vehicles file claims

2026:

- 10% of **4-year tenure policies**
- Claims distributed evenly from:
  Jan 1 – Feb 28